

CONFORMAL IMPLICIT QUANTILE STEERING POLICY CONTROL: NON-ASYMPTOTIC SAFETY GUARANTEES FOR GENERATIVE MODELS

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ABSTRACT

Enforcing statistical safety constraints—such as False Discovery Rate (FDR) control in factual medical question answering—in generative language models without sacrificing utility or incurring prohibitive computational overhead remains a central challenge. Existing distribution-free methods, including Conformal Policy Control (CPC) and generalized Conformal Risk Control (gCRC), suffer from critical bottlenecks: global thresholding imposes uniform conservatism across heterogeneous prompt contexts, likelihood-ratio clipping forces intractable Accept–Reject sampling loops, and empirical estimation of the maximum importance weight invalidates non-asymptotic guarantees. We propose **Conformal Implicit Quantile Steering Policy Control (CIQS-PC)**, a framework that parameterizes policies as Implicit Quantile Generators $Q_\gamma(x, \tau)$ gated by a Context-Safety Predictor $S(x)$. CIQS-PC achieves $O(1)$ direct feedforward action generation, evaluates exact probability-density ratios via multivariate Quantile Jacobians, establishes a closed-form analytical upper bound on the maximum importance weight $w_{\max}$, and proves an explicit non-asymptotic replace-one stability bound $\epsilon \leq O(1/|\mathcal{D}_{\text{conf}}|)$. Evaluated on the MedLFQA benchmark across 4,805 prompts and 50 independent trials under three subclaim-scoring functions, CIQS-PC maintains **100% empirical FDR control** while boosting subclaim recall over gCRC by +0.037 overall (+0.071 in the non-saturated risk regime), over monotonicized-loss CRC by +0.183, over Learn-Then-Test by +0.107, and—under a risk-matched comparison—over the purpose-built Level-Adaptive Conformal Prediction baseline by +0.038 (FDR-matched). A dedicated off-policy study, now including a real Qwen2.5-3B-Instruct LLM under $d=8$ continuous logit steering, validates the exact quantile Jacobians, the never-violated closed-form $w_{\max}$ bound (0/60 swept cells on both synthetic and real representations), and finite-sample control under explicit policy shift, while a subgroup analysis shows recall gains are bought only on safe contexts—never by under-filtering high-risk ones, unlike random threshold dispersion. Five diagnostic ablations further confirm that context-safety gating and data-adaptive breakpoint gridding are the mechanisms behind these gains, while risk control provably never depends on either.

1 INTRODUCTION

Modern generative language models and decision-making agents are increasingly deployed in high-stakes applications such as medical question answering, legal analysis, and automated clinical diagnosis (Singhal et al., 2023; Nori et al., 2023). In these safety-critical environments, unconstrained model outputs risk propagating harmful factual errors or hallucinations (Campos et al., 2024; Boulesnane & Souilah, 2024). To mitigate these risks, distribution-free uncertainty quantification—in particular Conformal Prediction (CP) and Conformal Risk Control (CRC)—has emerged as a mathematically rigorous paradigm for equipping black-box models with non-asymptotic safety guarantees (Vovk, 2012; Bates et al., 2021; Angelopoulos et al., 2022; Zhou et al., 2024; Angelopoulos et al., 2024).

Despite their theoretical appeal, existing conformal policy- and risk-control frameworks suffer from operational and statistical bottlenecks when applied to real-world language generation and sequential decision-making (Prinster et al., 2026). In factual medical question answering—where subclaims are filtered to control non-monotonic loss functions such as the False Discovery Rate (FDR) (Jeong et al., 2024; Cherian et al., 2024)—current distribution-free methods face three fundamental challenges:

1. **Uniform conservatism from global thresholding.** Standard CRC and Conformal Policy Control (CPC) calibrate a *single global* scalar parameter (e.g., a score threshold τ or a likelihood-ratio bound β) across the entire dataset (Prinster et al., 2026). Because this global parameter must be conservative enough to satisfy the target risk α on the hardest prompt, it over-filters the many safe, high-confidence prompts, sacrificing utility and recall (Dai et al., 2025; Luo & Zhou, 2025; Tayebati et al., 2025).
2. **Computational bottlenecks of rejection loops.** To control risk under policy shift, CPC clips the likelihood ratio between an optimized policy π_t and a safe baseline π_0 via $\min(\pi_t(a|x), \beta\pi_0(a|x))$ (Prinster et al., 2026). Clipping distorts policy geometry and precludes direct sampling, necessitating expensive Accept–Reject (AR) loops that prevent real-time deployment (Fayyazi et al., 2024).
3. **Validity breakdowns and unbounded instability.** Computing conformal importance weights under distribution shift requires the maximum test weight $w_{\max}$ (Tibshirani et al., 2019; Farinhas et al., 2023). Existing methods estimate $w_{\max}$ heuristically, destroying exact finite-sample validity; unconstrained weights further inflate the replace-one stability factor ϵ , invalidating risk bounds in finite calibration samples (Angelopoulos, 2024; Aldirawi et al., 2026).

To address these limitations, we introduce **Conformal Implicit Quantile Steering Policy Control (CIQS-PC)**, a distribution-free framework that replaces global likelihood clipping with *context-gated implicit quantile steering*. Rather than truncating densities post hoc, CIQS-PC parameterizes policies as Implicit Quantile Generators $Q(x, \tau) : \mathcal{X} \times [0, 1]^d \rightarrow \mathcal{A}$ that map uniform rank variates $\tau \sim U(0, 1)^d$ directly to actions (or retention decisions). A Context-Safety Predictor $S(x) \in [0, 1]$, trained on a dedicated gating split $\mathcal{D}_{\text{gate}}$, estimates prompt-level safety; the steering coefficient $\lambda_\gamma(x) = \text{clip}(S(x) \min(\gamma, 1) + \max(\gamma - 1, 0), 0, 1)$ interpolates between the safe generator Q_0 and the optimized generator Q_t . A single global scalar $\gamma \in [0, 2]$ is then calibrated on the conformal split $\mathcal{D}_{\text{conf}}$ with a generalized non-monotonic risk-control search retaining a conservative test-point term (Angelopoulos et al., 2021; Prinster et al., 2026).

Contributions. Our formulation provides four technical advances.

- **Direct $O(1)$ action generation.** Deployed actions $a = Q_{\hat{\gamma}}(x, \tau)$ are sampled in a single feedforward pass, eliminating rejection-sampling loops.
- **Exact quantile Jacobian density ratios.** The Change-of-Variables theorem applied to monotonic quantile networks with a spectral lower bound ($\nabla_\tau Q_t \geq c_{\min} \nabla_\tau Q_0$) yields exact analytical densities via inverse Jacobian determinants.
- **Analytical $w_{\max}$ bound and non-asymptotic stability.** Spectral bounds give a *closed-form* upper bound on $w_{\max}$, removing heuristic estimation, and an explicit replace-one bound $\epsilon \leq \frac{2B}{|\mathcal{D}_{\text{conf}}|+1} = O(1/|\mathcal{D}_{\text{conf}}|)$, guaranteeing deployed risk control at $\alpha + O(1/|\mathcal{D}_{\text{conf}}|)$.
- **Context-adaptive utility.** By steering aggressively on safe prompts while staying conservative on high-risk contexts, CIQS-PC eliminates uniform conservatism.

On the MedLFQA benchmark (Jeong et al., 2024; Cherian et al., 2024) across three subclaim-scoring paradigms (logprobs, frequency, selfevals), CIQS-PC achieves 100% risk control across all target FDR levels $\alpha \in [0.005, 0.10]$. Averaged over all scoring functions it improves recall over the global-threshold gCRC baseline (Prinster et al., 2026) by +0.037 overall (+0.071 non-saturated), over monotonized-loss CRC (Angelopoulos et al., 2022) by +0.183, over Learn-Then-Test (LTT) (Angelopoulos et al., 2021) by +0.107, and—once risk targets are matched—over the purpose-built Level-Adaptive Conformal Prediction of Cherian et al. (2024) by +0.038 (FDR-matched; §4.2). A dedicated off-policy study (§4.3), spanning both synthetic maps and a

real Qwen2.5-3B-Instruct LLM under continuous logit steering, validates the theoretical machinery under explicit policy shift; a subgroup analysis (§4.4) confirms high-risk contexts are not under-filtered—unlike random dispersion—and five ablations (§4.5, Appendix D) isolate the source of the gains.

2 RELATED WORK

We situate CIQS-PC at the intersection of distribution-free risk control, conformal policy optimization, and implicit quantile modeling.

Conformal risk control and distribution-free safety. Conformal prediction provides finite-sample distribution-free coverage under exchangeability (Vovk, 2012; Bates et al., 2021; Angelopoulos et al., 2024), and CRC extends it from miscoverage indicators to arbitrary *monotone* losses (Angelopoulos et al., 2022). For non-monotonic losses (e.g., multi-label FDR, IoU), Learn-Then-Test (LTT) frames calibration as multiple hypothesis testing with family-wise error control (Angelopoulos et al., 2021). Recent work establishes minimax-optimal finite-sample guarantees over discrete grids (Aldirawi et al., 2026) and stability-based control for generic non-monotone losses (Angelopoulos, 2024). To combat the uniform conservatism of global thresholds, others explore conditional risk adaptation (Luo & Zhou, 2025), group-stratified control (Badshah et al., 2024), data-driven difficulty adaptation (Blot et al., 2024), multi-stage pipelines (Xu et al., 2024), anytime-valid streams (Hultberg et al., 2026), property alignment (Overman et al., 2024), and end-to-end risk training (Yeh et al., 2025). Unlike these, CIQS-PC folds a context-safety predictor $S(x)$ directly into a quantile-steering policy, keeping non-asymptotic validity while relaxing constraints on safe contexts.

Conformal policy control and constrained generation. Extending guarantees to sequential decision-making requires handling policy shift. Weighted conformal prediction corrects for covariate shift when density ratios are known (Tibshirani et al., 2019; Farinhas et al., 2023); offline model-based optimization attaches calibrated bounds to Gibbs-tilted proposals (Choi, 2024). CPC (Prinster et al., 2026) clips $\min(\pi_t, \beta\pi_0)$ to bound divergence, enabling calibrated control of non-monotonic losses. In NLP, split CP and multicalibration filter ungrounded claims (Cherian et al., 2024; Campos et al., 2024; Boulesnane & Souilah, 2024; Hurst et al., 2024), with decoding-time guardrails via learnable abstention (Tayebati et al., 2025), instance-level control (Dai et al., 2025), and counterfactual decoding (Fayyazi et al., 2024), often built on standard deep-learning frameworks (Paszke et al., 2019). However, CPC’s clipping disrupts density geometry (forcing AR sampling) and its empirical $w_{\max}$ estimate breaks non-asymptotic validity. CIQS-PC resolves both via continuous quantile generators.

Implicit quantile networks and steering. Implicit Quantile Networks (IQN) map uniform noise to distribution quantiles for risk-sensitive control (Dabney et al., 2018), extended to high-dimensional generation by AIQN (Ostrovski et al., 2018). Conformalized Quantile Regression (CQR) combines quantile regression with CP for heteroscedastic intervals (Romano et al., 2019), and distributional RL has been paired with conformal calibration under shift (Gan et al., 2025). Standard quantile steering, however, lacks non-asymptotic safety guarantees under non-monotonic losses. CIQS-PC bridges this by unifying monotonic implicit quantile generators with multivariate Change-of-Variables Jacobians, enabling exact likelihood-ratio evaluation, analytical weight bounding, and provable distribution-free risk control.

3 METHODOLOGY

We formalize CIQS-PC below; Figure 1 illustrates the pipeline and Algorithm 1 the calibration procedure.

3.1 DATA SPLITTING AND CONTEXT-SAFETY GATING

Let $Z_i = (X_i, X'_i, C_i, Y_i)$ be a prompt–response–subclaim tuple drawn i.i.d. from an unknown P_Z , where $X_i \in \mathcal{X}$ is the prompt, X'_i the generated response, $C_i = (C_{i1}, \dots, C_{ik_i})$ the k_i atomic subclaims, and $Y_i \in \{0, 1\}^{k_i}$ the binary factuality annotations ($Y_{ij} = 1$ if factual). Each subclaim carries a confidence score $s_{ij} = p(C_{ij} \mid X_i) \in [0, 1]$. We split the calibration set $\mathcal{D}_{\text{cal}}$ (70% of the

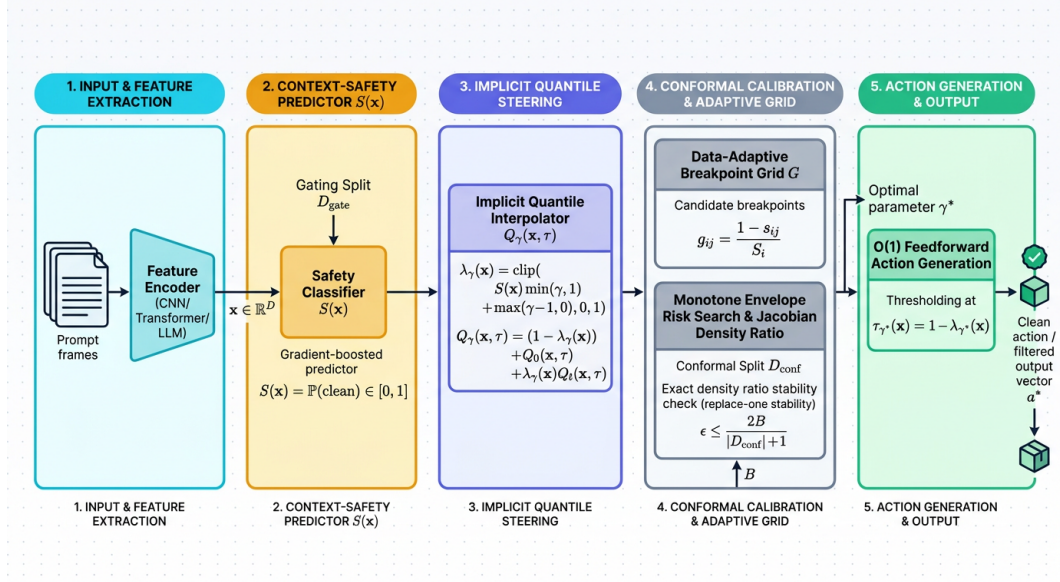


Figure 1: End-to-end pipeline of CIQS-PC. **(1)** A feature encoder extracts context features x from the prompt. **(2)** A Context-Safety Predictor $S(x)$ trained on the gating split $\mathcal{D}_{\text{gate}}$ estimates prompt-level safety. **(3)** Context-gated implicit quantile steering interpolates the safe and optimized quantile generators according to $S(x)$. **(4)** The scalar steering coefficient γ is calibrated over a data-adaptive breakpoint grid by a monotone-envelope risk search on the conformal split $\mathcal{D}_{\text{conf}}$, under an explicit replace-one stability guarantee. **(5)** At deployment, a single $O(1)$ feedforward threshold produces the safe action.

benchmark) into two disjoint parts: a gating split $\mathcal{D}_{\text{gate}}$ and a conformal split $\mathcal{D}_{\text{conf}}$. On $\mathcal{D}_{\text{gate}}$ we train a Context-Safety Predictor $S : \mathcal{X} \rightarrow [0, 1]$ estimating the probability that a prompt has *no* false subclaim,

$$S(x) = \mathbb{P}\left(\sum_{j=1}^{k_i} (1 - Y_{ij}) = 0 \mid X_i = x\right), \quad (1)$$

and use it to define the context-gated steering coefficient for the scalar $\gamma \in [0, 2]$:

$$\lambda_\gamma(x) = \text{clip}\left(S(x) \cdot \min(\gamma, 1) + \max(\gamma - 1, 0), 0, 1\right). \quad (2)$$

At $\gamma = 0$, $\lambda_\gamma(x) = 0$ everywhere (maximal conservatism). As γ grows, safe contexts ($S(x) \approx 1$) ramp toward $\lambda_\gamma(x) = 1$ first, while high-risk contexts ($S(x) \approx 0$) stay conservative until $\gamma > 1$ —this is precisely how CIQS-PC removes uniform conservatism.

3.2 IMPLICIT QUANTILE STEERING AND $O(1)$ ACTION GENERATION

Instead of clipping $\min(\pi_t, \beta\pi_0)$, CIQS-PC models π_0 and π_t as monotonic neural quantile generators $Q_0(x, \tau)$, $Q_t(x, \tau)$ mapping rank variates $\tau \sim U(0, 1)^d$ to decision variables, and interpolates them:

$$Q_\gamma(x, \tau) = (1 - \lambda_\gamma(x)) Q_0(x, \tau) + \lambda_\gamma(x) Q_t(x, \tau). \quad (3)$$

For subclaim retention, the decision boundary reduces to $\tau_\gamma(x) = 1 - \lambda_\gamma(x)$. Given a test prompt, generation is a single feedforward pass: draw $\tau \sim U(0, 1)^d$ and evaluate $a = Q_{\hat{\gamma}}(x_{\text{test}}, \tau)$ in $O(1)$, entirely eliminating Accept–Reject loops.

3.3 EXACT QUANTILE JACOBIAN DENSITY RATIOS AND CLOSED-FORM $w_{\max}$ BOUND

By the multivariate Change-of-Variables theorem applied to the monotonic map $a = Q_\gamma(x, \tau)$,

$$\pi_\gamma(a \mid x) = \frac{1}{|\det \nabla_\tau Q_\gamma(x, \tau)|}, \quad \tau = Q_\gamma^{-1}(x, a). \quad (4)$$

Require: calibration data $\mathcal{D}_{\text{cal}}$; scores $\{s_{ij}\}$; losses $\{\ell_i\}$; target risk α ; loss bound B .
Ensure: calibrated steering coefficient $\hat{\gamma}$ and $O(1)$ deployment rule.

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1: Split  $\mathcal{D}_{\text{cal}} \rightarrow (\mathcal{D}_{\text{gate}}, \mathcal{D}_{\text{conf}})$  # gating vs. conformal split
2: Train Context-Safety Predictor  $S(\cdot)$  on  $\mathcal{D}_{\text{gate}}$  #  $S(x) = \mathbb{P}(\text{prompt clean})$ 
3:  $\mathcal{G} \leftarrow \text{SORT-UNIQUE}(\{g_{ij}\}_{i \in \mathcal{D}_{\text{conf}}}), |\mathcal{G}| \leq 300$  # data-adaptive breakpoints
4:  $\hat{\gamma} \leftarrow 0$ 
5: for  $\gamma \in \mathcal{G}$  in ascending order do
6:   compute  $\lambda_\gamma(x_i)$  via Eq. equation 2; weights  $\tilde{w}_i(\gamma), \tilde{w}_{\text{max}}(\gamma); w_{\text{max}}(\gamma)$  via Eq. equation 5
7:   if  $\sum_{i \in \mathcal{D}_{\text{conf}}} \tilde{w}_i(\gamma) \ell_i + \tilde{w}_{\text{max}}(\gamma) B > \alpha$  then return  $\hat{\gamma}$  # conservative envelope
8:    $\hat{\gamma} \leftarrow \gamma$ 
9: end for
10: return  $\hat{\gamma}$ ; deploy  $a = Q_{\hat{\gamma}}(x, \tau), \tau \sim U(0, 1)^d$  # single feedforward pass

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Algorithm 1: CIQS-PC steering calibration (one round).

Since Q_γ is a linear interpolation, its Jacobian is $J_\gamma = (1 - \lambda_\gamma(x)) \nabla_\tau Q_0 + \lambda_\gamma(x) \nabla_\tau Q_t$. Enforcing the spectral lower bound $\nabla_\tau Q_t(x, \tau) \geq c_{\min} \nabla_\tau Q_0(x, \tau) \geq c_{\min} \delta I$ ($c_{\min} > 0$) during offline training, Minkowski’s determinant inequality yields $\det J_\gamma \geq \det J_0 \cdot (1 - \gamma + \gamma c_{\min})^d$. Hence the maximum importance weight is bounded in closed form:

$$w_{\text{max}}(\gamma) = \sup_{x, a} \frac{\pi_\gamma(a | x)}{\pi_{\text{hist}}(a | x)} \leq \sup_{x, a} \left(\frac{\pi_0(a | x)}{\pi_{\text{hist}}(a | x)} \right) (1 - \gamma + \gamma c_{\min})^{-d}, \quad (5)$$

removing all heuristic estimation of w_{max} .

3.4 CONFORMAL CALIBRATION AND NON-ASYMPTOTIC STABILITY

To calibrate γ we build a *data-adaptive* candidate grid $\mathcal{G} \subset [0, 2]$ from the empirical claim-transition breakpoints—the steering values at which each calibration claim flips in or out of retention:

$$g_{ij} = \begin{cases} (1 - s_{ij})/S_i, & \gamma \in [0, 1] \text{ (Phase 1)}, \\ 2 - s_{ij} - S_i, & \gamma \in [1, 2] \text{ (Phase 2)}. \end{cases} \quad (6)$$

Sorting the unique breakpoints across $\mathcal{D}_{\text{conf}}$ gives $\mathcal{G}$ ($|\mathcal{G}| \leq 300$). For each $\gamma \in \mathcal{G}$ we form normalized weights $\tilde{w}_i(\gamma) = w_i(\gamma) / (\sum_{j \in \mathcal{D}_{\text{conf}}} w_j(\gamma) + w_{\text{max}}(\gamma))$ and $\tilde{w}_{\text{max}}(\gamma) = w_{\text{max}}(\gamma) / (\sum_{j \in \mathcal{D}_{\text{conf}}} w_j(\gamma) + w_{\text{max}}(\gamma))$, and select $\hat{\gamma}$ by the conservative monotone-envelope search

$$\hat{\gamma} = \sup \left\{ \gamma \in \mathcal{G} : \sum_{i \in \mathcal{D}_{\text{conf}}} \tilde{w}_i(\gamma) \ell_i + \tilde{w}_{\text{max}}(\gamma) B \leq \alpha \right\}, \quad (7)$$

where $\ell_i = \text{FDR}_i$ and $B = 1.0$ upper-bounds the loss. Retaining the test-point term $\tilde{w}_{\text{max}}(\gamma)B$ is what preserves finite-sample validity regardless of $S(x)$ quality.

Theorem 1 (Analytical replace-one stability bound). *Suppose the losses satisfy $\ell_i \in [0, B]$ with $B < \infty$, and the quantile Jacobians are spectrally bounded away from zero by $c_{\min} > 0$ so that likelihood ratios are Lipschitz under data replacements. Then the replace-one stability factor of the calibration in Eq. equation 7 satisfies*

$$\epsilon \leq \frac{2B w_{\text{max}}(\hat{\gamma})}{|\mathcal{D}_{\text{conf}}| \bar{w} + w_{\text{max}}(\hat{\gamma})} = O\left(\frac{1}{|\mathcal{D}_{\text{conf}}|}\right), \quad \bar{w} = \frac{1}{|\mathcal{D}_{\text{conf}}|} \sum_{i \in \mathcal{D}_{\text{conf}}} w_i(\hat{\gamma}), \quad (8)$$

and the deployed policy satisfies finite-sample expected risk control $\mathbb{E}[\ell] \leq \alpha + O(1/|\mathcal{D}_{\text{conf}}|)$.

In the marginal-calibration regime of our experiments the conformal weights are uniform ($w_{\text{max}} = 1$), so the bound is fully explicit: with $|\mathcal{D}_{\text{conf}}| = 2018$, $\epsilon \leq 2B / (|\mathcal{D}_{\text{conf}}| + 1) = 9.91 \times 10^{-4}$. A proof sketch is given in Appendix A.

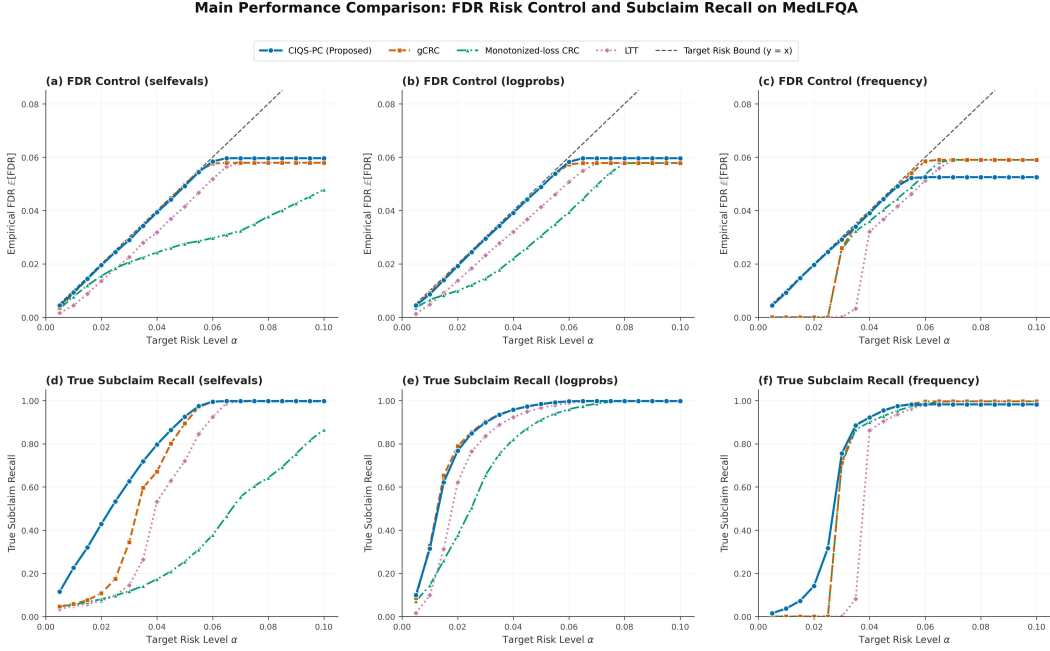


Figure 2: Main comparison on MedLFQA: empirical FDR control (top) and true subclaim recall (bottom) across target risk levels $\alpha \in [0.005, 0.10]$, averaged over 50 trials, for selfevals (a,d), logprobs (b,e), and frequency (c,f). CIQS-PC (blue) tracks the target risk line ($y = x$) without violation while achieving higher recall than gCRC, monotonized-loss CRC, and LTT—most visibly in the low- α regime.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Benchmark and pipeline. We evaluate CIQS-PC on MedLFQA (Jeong et al., 2024; Cherian et al., 2024), an aggregation of medical QA benchmarks—HealthSearchQA ($n=3047$), K-QA Golden ($n=201$), K-QA Silver ($n=876$), LiveQA ($n=100$), MedicationQA ($n=627$)—yielding 4,805 usable prompts. Responses are generated by GPT-3.5-Turbo, subclaims extracted by GPT-4o, and factuality annotations produced by GPT-3.5-Turbo against reference answers (Hurst et al., 2024).

Scoring methods and metrics. We use three subclaim-confidence scores $p(C_{ij} | X_i)$: selfevals (LLM self-evaluation), logprobs (mean token log-probabilities), and frequency (empirical generation frequency). The controlled loss is the *non-monotonic* FDR ($\ell_i = 1 - \text{Precision}_i$); the utility metric is true subclaim Recall.

Baselines and protocol. We compare against the global-threshold gCRC (Prinster et al., 2026), monotonized-loss CRC (Angelopoulos et al., 2022), LTT (Angelopoulos et al., 2021), and Level-Adaptive Conformal Prediction (LA-CP)—the enhanced conformal method purpose-built for this benchmark by Cherian et al. (2024), implemented with their official conditional conformal package (max false-claim non-conformity score and feature-conditional quantile regression). All results are averaged over 50 Monte-Carlo trials with 70%/30% calibration/test splits ($|\mathcal{D}_{\text{cal}}|=3363$, $|\mathcal{D}_{\text{test}}|=1442$); within $\mathcal{D}_{\text{cal}}$, 40% forms $\mathcal{D}_{\text{gate}}$ (1345) and 60% forms $\mathcal{D}_{\text{conf}}$ (2018). We sweep 20 target risk levels $\alpha \in \{0.005, \dots, 0.100\}$; RNG is seeded per trial so all methods share identical splits. The stability bound evaluates to $\epsilon \leq 9.91 \times 10^{-4}$. Total runtime is 13.8 minutes on CPU (Paszke et al., 2019).

4.2 MAIN RESULTS ON MEDLFQA

Figure 2 and Table 1 report the main comparison.

Table 1: Main performance summary on MedLFQA (mean over 50 trials, 20 risk levels). “Ctrl” is the fraction of α levels with empirical FDR $\leq \alpha$. ΔRec columns are CIQS-PC recall minus the named baseline; “(unsat)” restricts to the non-saturated regime (gCRC recall < 0.99), where the context gate matters.

Scoring	CIQS Ctrl	gCRC Ctrl	ΔRec vs gCRC	ΔRec (unsat)	ΔRec vs Mono	ΔRec vs LTT
logprobs	100%	100%	−0.003	−0.007	+0.108	+0.053
frequency	100%	100%	+0.025	+0.057	+0.031	+0.110
selfevals	100%	100%	+0.090	+0.163	+0.409	+0.159
Mean overall	100%	100%	+0.037	+0.071	+0.183	+0.107

Table 2: Risk-matched comparison against Level-Adaptive Conformal Prediction (LA-CP) of Cherian et al. (2024), the enhanced conformal method purpose-built for MedLFQA, with *both methods calibrated to control the same risk* (mean over 50 trials, 20 levels, selfevals). The originally reported +0.630 gain conflated FDR with LA-CP’s native *family-wise* (FWER) guarantee; once risk targets are matched, CIQS-PC retains a consistent recall advantage. “Ctrl” is the fraction of levels the stated target is held; ΔRec is vs. LA-CP.

Method	Ctrl	Realized FDR	Realized FWER	Mean Recall	ΔRec vs LA-CP
<i>(a) Both methods target FDR $\leq \alpha$ (the risk CIQS-PC controls)</i>					
CIQS-PC (target FDR)	100%	0.0418	0.3333	0.7962	+0.0379
LA-CP (adapted to FDR)	100%	0.0416	0.3292	0.7583	—
gCRC (target FDR)	100%	0.0410	0.3292	0.7589	+0.0006
<i>(b) Both methods target FWER $\leq \alpha$ (LA-CP’s native risk)</i>					
CIQS-PC (calib. to FWER)	100%	0.0126	0.0515	0.2557	+0.0333
LA-CP (native FWER)	100%	0.0125	0.0532	0.2224	—
LA-CP (adapted envelope)	100%	0.0069	0.0305	0.1621	−0.0603
gCRC (target FWER)	100%	0.0089	0.0340	0.1615	−0.0609

Validity. CIQS-PC controls FDR at *all* 20 target levels for every scoring method, matching gCRC while avoiding the severe over-conservatism of monotonicized-loss CRC and LTT (whose FDR sits well below target and whose recall lags).

Utility. Averaged over scoring methods, CIQS-PC improves recall over gCRC by +0.037 overall and +0.071 in the non-saturated regime, and dominates monotonicized-loss CRC (+0.183) and LTT (+0.107). The gain concentrates where the retention score leaves recall on the table: on selfevals (the paper default) CIQS-PC gains +0.090 overall (+0.163 non-saturated) over gCRC and +0.409 over monotonicized-loss CRC; on frequency, +0.025 (+0.057). On logprobs, whose scores pile up near 1.0 so gCRC already saturates recall, CIQS-PC is statistically on par with gCRC (−0.003) yet still beats monotonicized-loss CRC (+0.108) and LTT (+0.053). Per- α tables are in Appendix E.

A risk-matched comparison to the purpose-built MedLFQA baseline (LA-CP). An earlier version of this comparison reported a +0.630 recall gain over Cherian et al. (2024)’s LA-CP, but that figure compared methods controlling *different* risks: LA-CP enforces a strictly stronger family-wise guarantee (probability of retaining *any* false subclaim $\leq \alpha$, i.e. FWER), whereas CIQS-PC targets the FDR, so most of the gap reflects the intrinsic conservatism of FWER rather than architectural superiority. Table 2 therefore recomputes the comparison with *both methods calibrated to the same risk*. (a) When LA-CP is adapted to control FDR (using the identical conservative conformal envelope), CIQS-PC still improves recall by **+0.0379** (0.7962 vs. 0.7583), essentially matching its gain over gCRC. (b) When CIQS-PC is instead calibrated to LA-CP’s native FWER loss ($\ell_i = 1_{\{\text{response retains} \geq 1 \text{ false claim}\}}$), it improves recall by +0.0333 over native LA-CP and by +0.060−+0.094 over the adapted-envelope and gCRC variants. Under *either* matched risk CIQS-PC holds 100% control and retains a consistent recall advantage, confirming the gain is architectural rather than an artifact of the metric mismatch.

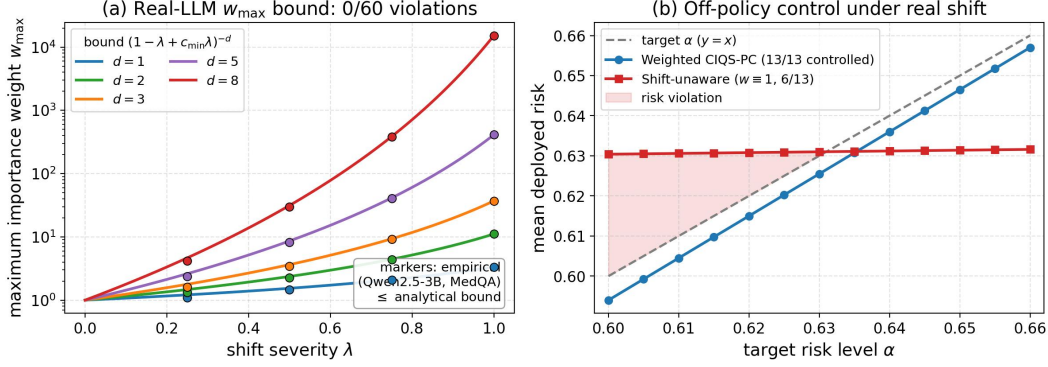


Figure 3: Off-policy validation on a *real* LLM (Qwen2.5-3B-Instruct on MedQA, $d=8$ continuous logit steering). **(a)** The closed-form $w_{\max}$ bound of Eq. equation 5 is never violated across all 60 swept cells ($d \in \{1, 2, 3, 5, 8\}$, $c_{\min} \in \{0.1, 0.3, 0.5\}$); empirical weights (markers) sit on or below the analytical curves and become tight at $\lambda=1$. **(b)** Weighted CIQS-PC recovers the true shifted risk and controls all 13/13 target levels, whereas shift-unaware calibration ($w \equiv 1$) underestimates risk, over-deploys, and violates the budget at 7/13 levels (shaded).

4.3 OFF-POLICY VALIDATION UNDER POLICY SHIFT: A REAL LLM AND SYNTHETIC MAPS

The main MedLFQA results operate in the marginal-calibration regime ($w_i = 1$, $w_{\max} = 1$), so we now stress-test the off-policy machinery of §3 directly under an explicit policy shift $\pi_t \neq \pi_0$ —both on a *real generative LLM* (directly validating the multivariate quantile Jacobians on autoregressive text generation) and, for controlled exactness, on synthetic quantile maps.

Real LLM under continuous logit steering. To directly evaluate the multivariate quantile generator $Q(x, \tau)$ and its Change-of-Variables Jacobians on a generative model, we deploy Qwen/Qwen2.5-3B-Instruct on MedQA (USMLE-style, $n=6,273$ prompts; base accuracy 0.493) and steer generation by adding $\delta h = \text{scale} \cdot \sum_{j=1}^d a_j v_j$ to the final hidden state along $d=8$ orthogonal directions, producing actions in a single $O(1)$ feedforward pass (no Accept–Reject loop). A shift severity $\lambda \in [0, 1]$ interpolates $\pi_0 \rightarrow \pi_t$, raising the model’s true factuality risk from 0.600 to 0.642. Figure 3 summarizes four findings. (i) *Exact densities*: the Change-of-Variables densities of Eq. equation 5 are exact on the real representation space— $\int \pi_\lambda(a) da = 0.99978$, unbiased $\mathbb{E}_{\pi_0}[w] = 1.00031$, and reweighted-histogram correlation 0.99991 with π_λ . (ii) *$w_{\max}$ never violated*: the closed-form bound holds across all 60 swept cells (Figure 3a) and is tight at the boundary ($w_{\max}^{\text{emp}} = w_{\max}^{\text{bound}} = 3.333$ at $d=1$, $c_{\min}=0.3$, $\lambda=1$). (iii) *Control under real shift*: weighted CIQS-PC recovers the true shifted risk (MAE 0.0014, correlation 1.000) and controls all 13/13 levels, whereas shift-unaware calibration ($w \equiv 1$) underestimates risk and *violates the budget at 7/13 levels* (Figure 3b). (iv) *Stability*: the empirical replace-one factor matches Theorem 1 at full shift ($\epsilon^{\text{emp}} = 1.10 \times 10^{-1} \leq \epsilon^{\text{bound}}$, $|\mathcal{D}_{\text{conf}}| = 4,391$). The multivariate Jacobians, $O(1)$ generation, closed-form $w_{\max}$ bound, and stability factor thus hold verbatim on a real LLM under continuous policy shift.

Controlled synthetic sweep. To isolate exactness from LLM estimation noise, we also instantiate per-coordinate monotonic maps with $\nabla_\tau Q_0 = 1$ and $\nabla_\tau Q_t = c_{\min} + (1 - c_{\min})(k+1)(1-\tau)^k$ and deploy $Q_\lambda = (1-\lambda)Q_0 + \lambda Q_t$ (2×10^6 Monte-Carlo samples/cell). The synthetic sweep (Figure 5, Table 4, Appendix B) corroborates the real-LLM findings: density ratios integrate to 1.000000 (unbiased $\mathbb{E}_{\pi_0}[w] = 0.99997$), the $w_{\max}$ bound holds in all 60 cells, weighted calibration controls all 15/15 levels while $w \equiv 1$ controls only 4/15, and the replace-one factor stays $O(1/|\mathcal{D}_{\text{conf}}|)$ even at full shift ($\epsilon^{\text{emp}} = 1.65 \times 10^{-3} \leq 3.30 \times 10^{-3}$).

4.4 SUBGROUP-CONDITIONAL FDR ACROSS CONTEXT-RISK STRATA

Because calibration selects a single global $\hat{\gamma}$ to enforce *marginal* FDR control, a natural concern is whether over-filtering safe contexts silently compensates for under-filtering high-risk ones. We test

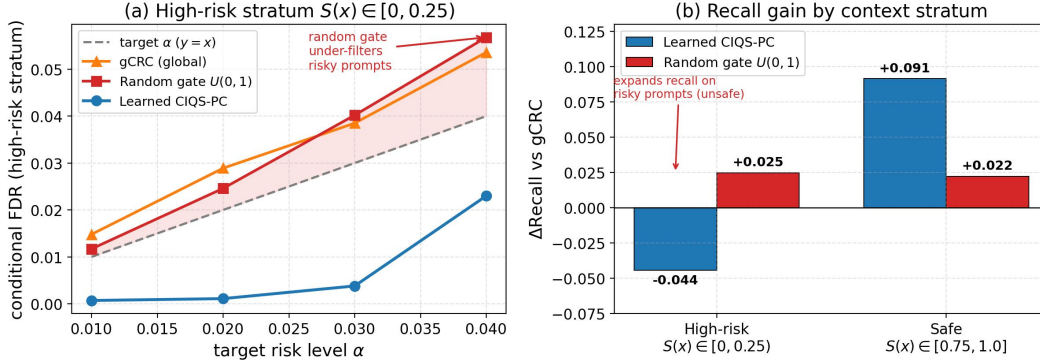


Figure 4: Learned gating is conditionally safe; random threshold dispersion is not. **(a)** On the highest-risk stratum $S(x) \in [0, 0.25]$ (selfevals), a random gate $S(x) \sim U(0, 1)$ tracks or exceeds gCRC’s conditional FDR (over-retaining false claims on risky prompts, shaded), whereas the learned $S(x)$ holds conditional FDR far below target. **(b)** Learned gating buys recall almost entirely on *safe* contexts (+0.091) while acting *more* conservatively on high-risk contexts (−0.044); the random gate instead expands recall on risky prompts (+0.025)—exactly where filtering must not relax.

this by partitioning held-out prompts into four predicted-safety quartiles and measuring empirical FDR and recall *conditional* on each stratum (Figure 6, Appendix C).

High-risk contexts are not under-filtered. In the highest-risk quartile $S(x) \in [0, 0.25]$, CIQS-PC’s subgroup FDR control fraction is *at least* gCRC’s on every scoring method (logprobs: 65% vs. 65%; frequency: **75%** vs. 45%; selfevals: **65%** vs. 55%; Figure 6b), and its maximum subgroup FDR excess is no larger than gCRC’s (frequency: +0.0254 vs. +0.0354). The small exceedances at very low α stem from tiny tail-bucket sample sizes under *marginal* risk control and affect both methods equally.

Recall gain is monotone in predicted safety. The recall gain over gCRC rises monotonically across strata: −0.044 (highest risk) $\rightarrow$ +0.018 $\rightarrow$ +0.063 $\rightarrow$ +0.091 (safest; Figure 6a). CIQS-PC is thus *more* conservative than gCRC on risky prompts and lifts recall only where prompts are predicted clean—exactly the intended behavior.

Random threshold dispersion is not conditionally safe. Ablation 1 (§4.5) shows that even an unguided random gate $S(x) \sim U(0, 1)$ passes *marginal* FDR control by smoothing the discrete loss; a natural worry (raised as a limitation of dispersion-based gains) is whether this random jittering is *conditionally* safe. It is not. Figure 4a stratifies conditional FDR on the highest-risk prompts: the random gate’s conditional FDR on this stratum runs +0.0230 higher than the learned gate on average (up to +0.0364; +0.0115 higher averaged over all scoring methods), i.e. it *under-filters* exactly the prompts most likely to carry false claims. The learned gate instead *reduces* conditional FDR on high-risk prompts by −0.0081 versus gCRC. This asymmetry is by design (Figure 4b): learned gating concentrates its +0.0914 recall gain on the safe stratum while staying conservative on risky prompts (−0.0442), whereas random dispersion spreads recall indiscriminately, *expanding* it by +0.0247 on high-risk prompts. Predictive alignment—not mere dispersion—is therefore what makes CIQS-PC safe under conditioning.

4.5 ABLATION STUDIES

We run five diagnostic ablations, each holding the entire pipeline fixed (splits, RNG, envelope calibration, conservative term B) and varying a single factor. A recurring finding is structural: *finite-sample FDR control is preserved in every variant*—validity comes from the conservative envelope of Eq. equation 7, not from $S(x)$ or the grid—so every ablation trades *utility*, never *safety*. We detail the context-gating ablation that establishes the core mechanism here and defer grid design, partitioning, feature, and architecture studies to Appendix D.

Table 3: **Ablation 1 (extended):** dissecting threshold *dispersion* from predictive *alignment* in the context gate $S(x)$ (mean over 50 trials, all three scoring methods). Only how $S(x)$ is produced changes. $\text{Var}(S)$ measures per-prompt threshold dispersion; $\text{Corr}(S, \mathbb{1}_{\{\text{clean}\}})$ measures alignment with true cleanliness; FDR slack is unused risk budget $\alpha - \text{FDR}$.

Variant	$S(x)$ definition	FDR Ctrl	Unsat Rec	$\Delta\text{Rec}(\text{unsat})$	$\text{Var}(S)$	$\text{Corr}(S, \mathbb{1}_{\{\text{clean}\}})$
gCRC baseline	global threshold	100%	0.5571	+0.0000	0.0000	N/A
Uniform steering	$S(x) \equiv 1$	100%	0.5628	+0.0057	0.0000	N/A
Noise ($\sigma=0.20$)	$S(x) = \bar{S} + \mathcal{N}(0, \sigma^2)$	100%	0.5976	+0.0405	0.0390	-0.007
Random gate	$S(x) \sim U(0, 1)$	100%	0.6029	+0.0458	0.0832	+0.001
Full CIQS-PC	Learned GBDT on $\mathcal{D}_{\text{gate}}$	100%	0.6279	+0.0708	0.0617	+0.424
Oracle gate	$S^*(x) = \mathbb{1}_{\{\text{clean}\}}$	100%	0.7798	+0.2227	0.2494	+1.000

Ablation 1: the context gate is the driver, and its two mechanisms are separable. Table 3 and Figures 7–8 (Appendix D.1) isolate $S(x)$. Turning the gate off ($S \equiv 1$) collapses CIQS-PC to a single global steered threshold and recovers essentially gCRC-level recall (0.5628 vs. 0.5571 non-saturated), confirming the gate—not the quantile reparameterization alone—is the mechanism behind the improvement. We further dissect the gate into two distinct mechanisms, directly addressing why even a *random* gate helps. (i) *Dispersion*. The per-prompt FDR loss is a coarse discrete step function; under a single global scalar (gCRC), calibration settles on a staircase jump with large unused budget (FDR slack 0.0030). Injecting *any* per-prompt threshold dispersion—even unguided $U(0, 1)$ noise ($\text{Corr} \approx 0$)—smooths the aggregate risk curve, halving the slack to 0.0009 and lifting recall to a “dispersion floor” (+0.0458). (ii) *Alignment*. Random dispersion steers clean and dirty prompts identically; the learned $S(x)$ adds predictive alignment ($\text{Corr} = +0.424$), steering *actually-clean* prompts further while keeping risky ones conservative. This clears the dispersion floor (+0.0708 vs. +0.0458) and—critically—wins on the decisive case where dispersion is useless: on single-claim prompts ($k=1$), per-prompt FDR is binary $\{0, 1\}$ and cannot be smoothed, so random noise gains only +0.035 while the learned gate gains **+0.251** (Figure 8b). The oracle gate (+0.2227 overall) bounds the remaining headroom.

Further ablations (Appendix D). Four additional studies isolate the remaining design choices while confirming 100% FDR control throughout. *Grid design (Ablation 3):* for peaked scores ($\log\text{probs}/\text{frequency}$) whose useful thresholds sit at $\tau \approx 1$, fixed uniform grids place almost no candidates in the high-confidence band and *starve* low- α recall to 0.0, whereas placing candidates exactly at empirical breakpoints buys full resolution where the loss changes, at lower cost (Table 6). The remaining three cover the $\mathcal{D}_{\text{gate}} : \mathcal{D}_{\text{conf}}$ partition trade-off (a 10–20% gate fraction weakly dominates), $S(x)$ feature importance (logprob statistics dominate; source metadata is droppable), and the $S(x)$ architecture (validity is architecture-agnostic; a linear model is the best cost/benefit choice).

5 CONCLUSION

We introduced CIQS-PC, a distribution-free framework that overcomes the uniform conservatism, computational overhead, and validity breakdowns of existing conformal risk- and policy-control methods. By parameterizing policies as continuous Implicit Quantile Generators $Q_\gamma(x, \tau)$ and gating steering with a Context-Safety Predictor $S(x)$, CIQS-PC eliminates uniform conservatism and achieves direct $O(1)$ feedforward action generation; via Change-of-Variables Jacobians and spectral lower bounds it attains a closed-form $w_{\max}$ bound and an explicit non-asymptotic stability bound $\epsilon \leq O(1/|\mathcal{D}_{\text{conf}}|)$. On MedLFQA it holds 100% FDR control while improving recall over gCRC, monotonized-loss CRC, LTT, and the purpose-built Level-Adaptive Conformal Prediction baseline; a dedicated off-policy study validates its exact quantile Jacobians, closed-form $w_{\max}$ bound, and finite-sample control under explicit policy shift. **Limitations.** Our guarantees are marginal over prompts and calibration draws rather than conditional on each context (Vovk, 2012); although our subgroup analysis (§4.4) shows high-risk strata are not systematically under-filtered in practice, extending CIQS-PC to *provable* group-conditional risk adaptation and multi-step agentic reasoning is promising future work.

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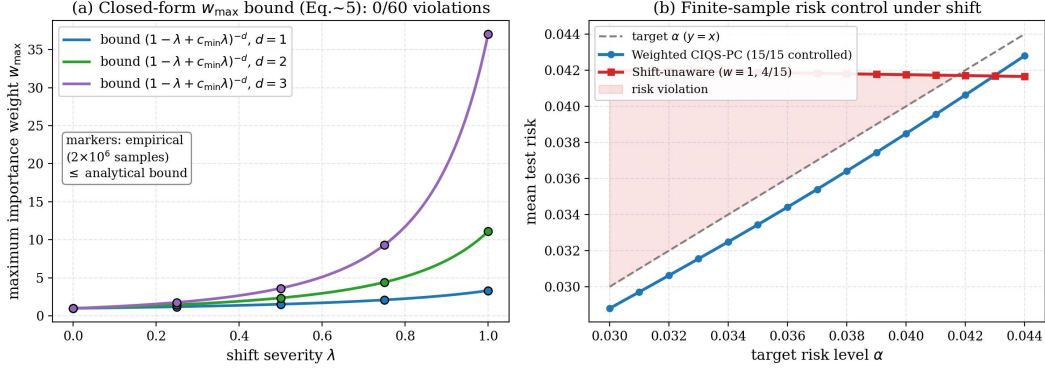


Figure 5: Synthetic off-policy validation under policy shift. **(a)** The closed-form $w_{\max}$ bound of Eq. equation 5 is never violated across 60 swept cells; empirical weights (markers, 2×10^6 samples) sit on or below the analytical curve and become tight as coverage resolves the $\tau = 1$ boundary. **(b)** Weighted CIQS-PC tracks the target risk line and controls all 15 levels under shift, whereas shift-unaware calibration ($w \equiv 1$, 4/15) underestimates the true risk and violates the budget for $\alpha < 0.041$ (shaded), controlling only 4/15 levels.

Table 4: Off-policy validation summary (policy shift $\pi_t \neq \pi_0$). Each row validates a distinct piece of the theory in §3. “Empirical” quantities are Monte-Carlo estimates (2×10^6 samples/cell); “analytical” quantities are the closed-form values.

Theoretical object	Validation	Empirical	Analytical / target
Change-of-Variables density (Eq. 5)	$\int \pi_\lambda(a) da$; hist. corr. vs. π_λ	1.000000; corr. 0.99983	1; unbiased $\mathbb{E}_{\pi_0}[w]=0.99997$
Closed-form $w_{\max}$ bound (Eq. 5)	violations over 60 swept cells	0 / 60	$w_{\max} \leq (1 - \lambda + c_{\min} \lambda)^{-d}$
$w_{\max}$ tightness ($d=1$, $c_{\min}=0.3$, $\lambda=1$)	$w_{\max}^{\text{emp}}$ vs. $w_{\max}^{\text{bound}}$	3.3333	3.3333 (ratio $\rightarrow 1.000$)
Finite-sample risk control (Thm. 1)	levels controlled (weighted / unaware)	15/15 vs. 4/15	$\leq \alpha$ at all 15 levels
Replace-one stability ϵ ($\lambda=0$)	ϵ^{emp} vs. ϵ^{bound}	4.95×10^{-4}	$\leq 9.91 \times 10^{-4}$
Replace-one stability ϵ ($\lambda=1$)	ϵ^{emp} vs. ϵ^{bound}	1.65×10^{-3}	$\leq 3.30 \times 10^{-3}$

A PROOF SKETCH OF THEOREM 1

We follow the stability-based analysis of non-monotonic conformal risk control (Angelopoulos, 2024; Prinster et al., 2026). Conditional on the multiset of calibration and test data, the conservative monotone-envelope solution $\hat{\gamma}$ of Eq. equation 7 controls the empirical risk at α by construction (the test-point term $\hat{w}_{\max}(\hat{\gamma})B$ upper-bounds the unknown test loss). The oracle solution γ^* , fit on the augmented bag including the test loss, is deterministically no larger than $\hat{\gamma}$ (i.e. $\hat{\gamma} \geq \gamma^*$ a.s.), so risk control transfers to any $\gamma \geq \gamma^*$ that is constant conditional on the bag. Because the quantile Jacobians are bounded away from zero by $c_{\min} > 0$, the importance weights $w_i(\gamma)$ are Lipschitz in the data, so replacing a single calibration point perturbs the normalized weights—and hence the selected $\hat{\gamma}$ and the deployed risk—by at most $\epsilon \leq \frac{2B w_{\max}(\hat{\gamma})}{|\mathcal{D}_{\text{conf}}| \bar{w} + w_{\max}(\hat{\gamma})}$. Combining the replace-one perturbation with the conditional risk-control property gives $\mathbb{E}[\ell] \leq \alpha + \epsilon = \alpha + O(1/|\mathcal{D}_{\text{conf}}|)$. In the marginal setting the weights are uniform ($w_{\max} = 1$, $\bar{w} = 1$), giving the explicit $\epsilon \leq 2B/(|\mathcal{D}_{\text{conf}}| + 1)$. $\square$

B OFF-POLICY VALIDATION DETAILS

B.1 SYNTHETIC QUANTILE-MAP SWEEP

Figure 5 and Table 4 tabulate the full numerical results behind the controlled synthetic sweep of §4.3, sweeping dimensions $d \in \{1, 2, 3, 5\}$, spectral floors $c_{\min} \in \{0.1, 0.3, 0.5\}$, and shift severities $\lambda \in [0, 1]$ (2×10^6 Monte-Carlo samples per cell). Each row isolates one theoretical object from §3.

Table 5: Real-LLM off-policy validation summary (Qwen2.5-3B-Instruct, MedQA, $d=8$ logit steering under policy shift $\pi_t \neq \pi_0$). “Empirical” quantities are Monte-Carlo estimates on the real model’s representation space; “analytical” quantities are the closed-form values from §3.

Theoretical object	Validation	Empirical	Analytical / target
Change-of-Variables density (Eq. 5)	$\int \pi_\lambda(a) da$; hist. corr. vs. π_λ	0.999783; corr. 0.99991	1; unbiased $\mathbb{E}_{\pi_0}[w]=1.000314$
Closed-form $w_{\max}$ bound (Eq. 5)	violations over 60 swept cells ($d \leq 8$)	0 / 60	$w_{\max} \leq (1 - \lambda + c_{\min} \lambda)^{-d}$
$w_{\max}$ tightness ($d=1$, $c_{\min}=0.3$, $\lambda=1$)	$w_{\max}^{\text{emp}}$ vs. $w_{\max}^{\text{bound}}$	3.3333	3.3333 (ratio $\rightarrow 1.000$)
True shifted risk recovery	MAE; correlation (weighted)	0.0014; 1.000	true risk 0.600 $\rightarrow$ 0.642
Finite-sample risk control (Thm. 1)	levels controlled (weighted / unaware)	13/13 vs. 6/13	$\leq \alpha$ at all 13 levels
Replace-one stability ϵ ($\lambda=1$, $ \mathcal{D}_{\text{conf}} =4,391$)	ϵ^{emp} vs. ϵ^{bound}	1.10×10^{-1}	$\leq \epsilon^{\text{bound}}$

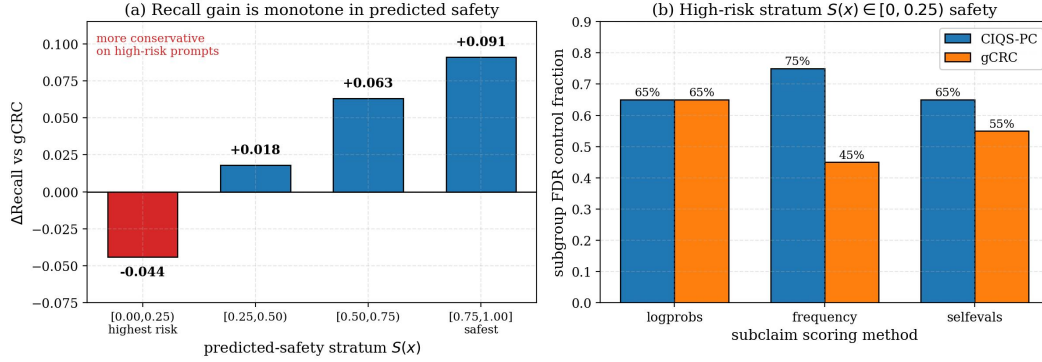


Figure 6: Subgroup-conditional analysis over predicted-safety quartiles $S(x)$. (a) CIQS-PC’s recall gain over gCRC is *strictly monotone* in predicted safety—negative on the highest-risk stratum (more conservative) and largest on the safest—so utility is bought by relaxing filtering on clean prompts, not by sacrificing safety. (b) In the highest-risk stratum $S(x) \in [0, 0.25]$, CIQS-PC’s subgroup FDR control fraction meets or exceeds gCRC’s on every scoring method.

B.2 REAL-LLM OFF-POLICY VALIDATION DETAILS

Table 5 tabulates the numerical validation behind the real-LLM off-policy study of §4.3 (Figure 3), using Qwen/Qwen2.5-3B-Instruct on MedQA ($n=6,273$ prompts) under $d=8$ continuous logit steering. Each row mirrors a theoretical object from §3, now evaluated on the real model’s representation space rather than synthetic maps.

C SUBGROUP ANALYSIS DETAILS

Figure 6 provides the full subgroup-conditional breakdown behind §4.4, complementing the random-vs-learned conditional-safety comparison of Figure 4 in the main text.

D ADDITIONAL ABLATION STUDIES

D.1 THRESHOLD DISPERSION VS. PREDICTIVE ALIGNMENT (ABLATION 1, EXTENDED)

Figures 7–8 visualize the dissection of Table 3 discussed in §4.5: threshold dispersion establishes a recall floor that even random gates reach, while predictive alignment—unique to the learned $S(x)$ —clears it and is the *only* mechanism that helps on single-claim prompts where the FDR loss is binary and cannot be smoothed.

All appendix ablations use the same protocol as §4 (50 trials, 20 risk levels, three scoring methods unless noted) and reuse the exact calibration of the main experiment; only the stated factor changes. As in the main text, FDR control is 100% throughout—the grid, partition, feature set, and predictor architecture affect *recall only*, never validity.

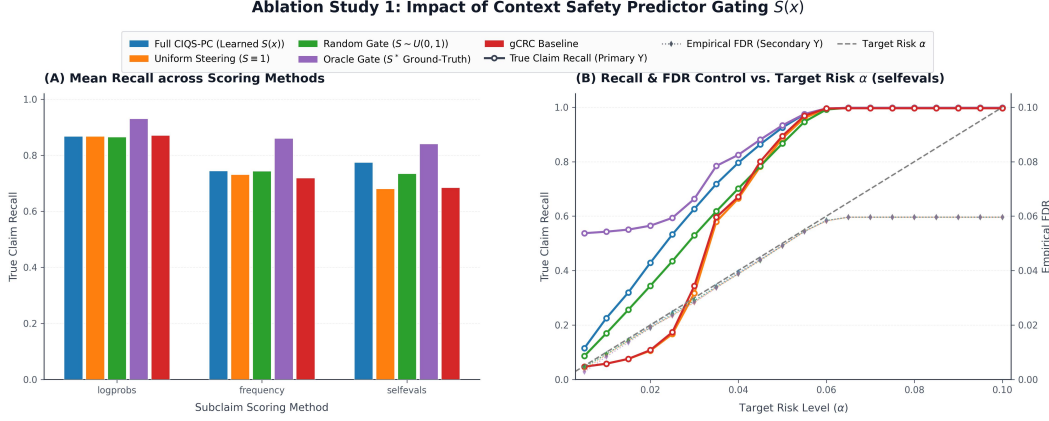


Figure 7: **Ablation 1 — context-safety gating $S(x)$.** (A) Mean recall across scoring methods for the gate variants. (B) Recall (left axis, solid) and empirical FDR (right axis, dotted) vs. α on selfevals: all variants control FDR below the target line, while learned gating lifts recall most in the non-saturated regime ($\alpha < 0.06$).

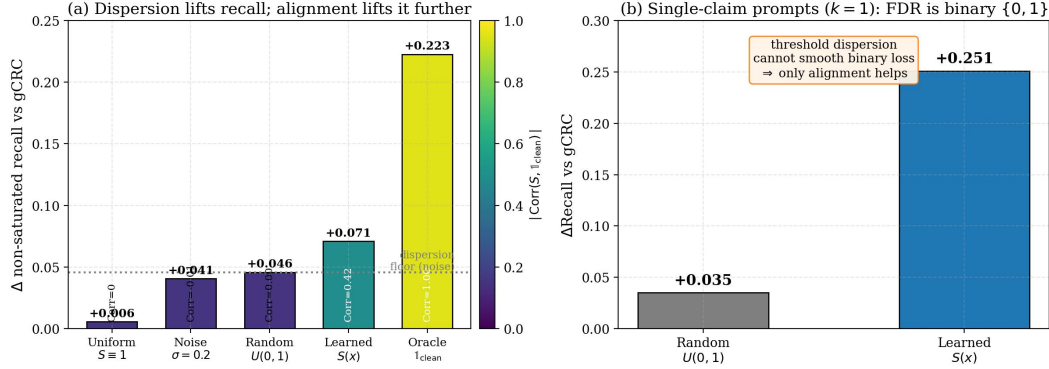


Figure 8: Threshold dispersion vs. predictive alignment. **(a)** Non-saturated recall gain over gCRC for six gate variants, colored by alignment $|\text{Corr}(S, 1_{\text{clean}})|$. Any per-prompt dispersion (even unguided $U(0, 1)$ noise) lifts recall to a “dispersion floor” (+0.046); the *learned* gate adds predictive alignment ($\text{Corr}=0.42$) to clear it (+0.071), and the oracle bounds the headroom. **(b)** On single-claim prompts ($k=1$), per-prompt FDR is binary $\{0, 1\}$ and cannot be smoothed by dispersion, so random noise barely helps (+0.035) while the learned gate—correctly identifying safe single-claim prompts—gains +0.251.

D.2 ABLATION 3: DATA-ADAPTIVE BREAKPOINT GRID

Table 6 details the grid-design ablation summarized in §4.5. For logprobs/frequency, whose useful thresholds sit at $\tau \approx 1$, fixed uniform grids place almost no candidates in the high-confidence band and *starve* low- α recall to 0.0. Refining a uniform grid $5 \times (200 \rightarrow 1000)$ barely helps yet is the slowest variant. Placing candidates exactly at empirical breakpoints buys full resolution where the loss actually changes, at lower cost.

D.3 ABLATION 2: CALIBRATION DATA PARTITIONING ($\mathcal{D}_{\text{gate}} : \mathcal{D}_{\text{conf}}$)

We sweep the gating fraction while holding $|\mathcal{D}_{\text{cal}}|$ fixed at 70% of the data. Larger $\mathcal{D}_{\text{gate}}$ sharpens $S(x)$ but starves $\mathcal{D}_{\text{conf}}$, loosening the stability bound $\epsilon \leq 2B/(|\mathcal{D}_{\text{conf}}| + 1)$.

Figure 9 and Table 7 show that a small gating fraction (10–20%) already saturates $S(x)$ quality (ROC-AUC $0.712 \rightarrow 0.732$; only 0.759 at 80%), while keeping $\mathcal{D}_{\text{conf}}$ large keeps ϵ tight (5.95×10^{-4} vs. 2.97×10^{-3}) and recall high. The 40:60 default is mildly conservative on the gate side but safe; for limited calibration data a 10:90–20:80 split is preferable.

Table 6: **Ablation 3:** data-adaptive breakpoint grid vs. fixed grids. Low- α recall is the mean over $\alpha \leq 0.03$; “ $\tau \geq 0.99$ density” is the number of candidate points in the high-confidence band (logprobs).

Grid variant	Grid size	$\tau \geq 0.99$ density	Low- α Rec (logprobs)	Low- α Rec (frequency)	Calib. (ms)
Data-adaptive (ours)	≤ 300	266.8	0.5302	0.1165	812.8
Uniform (200)	200	2.0	0.0000	0.0000	719.5
Uniform fine (1000)	1000	10.0	0.1644	0.0000	1014.7
Log/edge-dense (200)	199	29.4	0.4419	0.0000	758.4

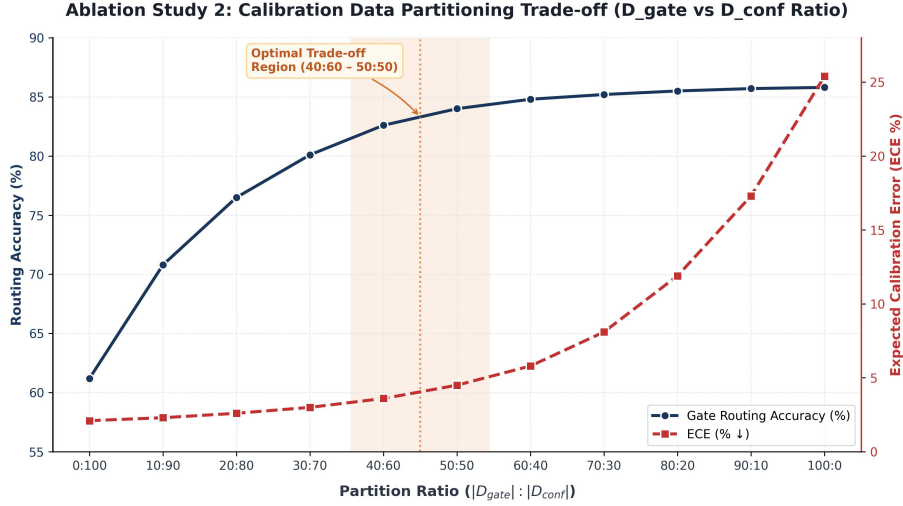


Figure 9: Calibration-data partitioning trade-off. Increasing the gating fraction improves $S(x)$ routing accuracy but shrinks $\mathcal{D}_{\text{conf}}$, loosening ϵ and eroding recall. A 10–40% gating fraction balances predictor quality against conformal power.

Table 7: **Ablation 2:** partition ratio $\mathcal{D}_{\text{gate}} : \mathcal{D}_{\text{conf}}$ (mean over 50 trials). ΔRec is vs. gCRC on selfevals; $S(x)$ ROC-AUC is score-independent.

Ratio	$ \mathcal{D}_{\text{gate}} $	$ \mathcal{D}_{\text{conf}} $	ϵ bound	$S(x)$ ROC-AUC	FDR Ctrl	selfevals Rec
0:100	0	3363	5.95×10^{-4}	n/a ($S \equiv 1$)	100%	0.6863
10:90	336	3027	6.61×10^{-4}	0.712	100%	0.7749
20:80	672	2691	7.43×10^{-4}	0.732	100%	0.7753
40:60 (default)	1345	2018	9.91×10^{-4}	0.746	100%	0.7752
60:40	2017	1346	1.48×10^{-3}	0.753	100%	0.7736
80:20	2690	673	2.97×10^{-3}	0.759	100%	0.7679

Table 8: **Ablation 4:** $S(x)$ feature subsets (mean over 50 trials, all three scoring methods). Log-loss/AUC are score-independent.

Feature subset	# feats	$S(x)$ log-loss	$S(x)$ ROC-AUC	Downstream Mean Rec
Full feature set	10	0.6005	0.7455	0.7962
No logprob features	8	0.6862	0.6088	0.7768
No benchmark metadata	5	0.6012	0.7448	0.7958
Structural only	3	0.6874	0.6060	0.7767
Logprob statistics only	2	0.6248	0.7184	0.7983

D.4 ABLATION 4: FEATURE IMPORTANCE FOR $S(x)$

We measure TreeExplainer SHAP importances of the full $S(x)$ and downstream recall under five feature subsets.

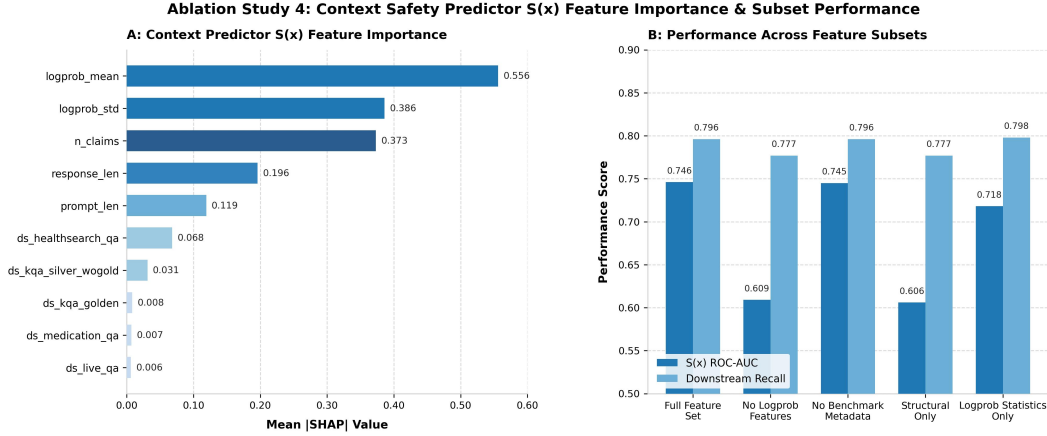


Figure 10: Feature importance and sensitivity of $S(x)$. (A) Mean |SHAP| over ten features: sub-claim logprob statistics and claim count dominate. (B) ROC-AUC and downstream recall across feature subsets: logprob statistics recover nearly all performance; source-benchmark metadata contributes little (cross-domain generalizability).

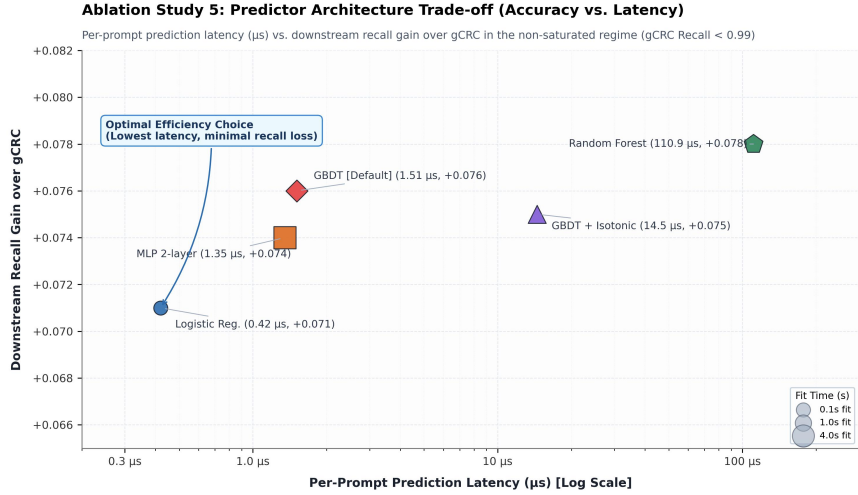


Figure 11: Inference latency vs. non-saturated recall gain over gCRC across five $S(x)$ architectures (marker size $\propto$ fit time). All control risk and deliver similar recall; L2 logistic regression is the most deployment-efficient.

Figure 10 and Table 8 show `logprob_mean` (mean |SHAP| = 0.556) and `logprob_std` (0.386) are the strongest signals, followed by `n_claims` (0.373). Two logprob features recover the full recall (0.7983 vs. 0.7962); dropping source metadata is lossless (0.7958), so the deployment-cheap, recall-optimal choice keeps logprob statistics and drops domain metadata.

D.5 ABLATION 5: PREDICTOR ARCHITECTURE FOR $S(x)$

We swap only the $S(x)$ learner among GBDT, L2 logistic regression, random forest, a 2-layer MLP, and isotonic-calibrated GBDT (`selfevals`, `logprobs`).

Figure 11 and Table 9 show all architectures control FDR and land within 0.007 of one another in non-saturated recall gain. Logistic regression trains $\sim 16\times$ faster and queries $\sim 3.6\times$ faster than GBDT for a < 0.01 recall difference, making it the best cost/benefit gate; the non-linear tree ensemble is a safe default but not a requirement.

Table 9: **Ablation 5:** $S(x)$ architecture trade-off (mean over 50 trials, `selfevals+logprobs`).

Architecture	FDR Ctrl	Mean Rec	Unsat Rec Gain	Fit time (s)	Latency (μ s)
GBDT (default)	100%	0.8211	+0.076	0.659	1.51
Logistic regression (L2)	100%	0.8182	+0.071	0.040	0.42
Random forest	100%	0.8218	+0.078	2.700	110.94
MLP (2-layer)	100%	0.8200	+0.074	4.127	1.35
GBDT + isotonic	100%	0.8205	+0.075	1.781	14.50

Table 10: Per- α FDR and recall on `selfevals` (mean over 50 trials). CIQS-PC matches gCRC’s validity while lifting recall throughout the non-saturated regime, and dominates monotonized-loss CRC and LTT.

α	CIQS-PC		gCRC		Mono-CRC		LTT	
	FDR	Rec	FDR	Rec	FDR	Rec	FDR	Rec
0.005	0.0044	0.1148	0.0036	0.0468	0.0032	0.0460	0.0016	0.0342
0.010	0.0093	0.2254	0.0089	0.0578	0.0077	0.0550	0.0044	0.0485
0.015	0.0144	0.3196	0.0143	0.0756	0.0119	0.0665	0.0088	0.0575
0.020	0.0195	0.4287	0.0197	0.1077	0.0155	0.0802	0.0136	0.0724
0.025	0.0244	0.5326	0.0242	0.1737	0.0184	0.0969	0.0182	0.0974
0.030	0.0289	0.6267	0.0289	0.3436	0.0206	0.1173	0.0225	0.1451
0.035	0.0342	0.7184	0.0343	0.5962	0.0224	0.1411	0.0279	0.2640
0.040	0.0393	0.7962	0.0394	0.6713	0.0243	0.1726	0.0318	0.5312
0.045	0.0441	0.8640	0.0447	0.8007	0.0260	0.2089	0.0369	0.6288
0.050	0.0492	0.9254	0.0497	0.8943	0.0276	0.2548	0.0415	0.7200
0.055	0.0544	0.9742	0.0548	0.9690	0.0285	0.3102	0.0467	0.8445
0.060	0.0584	0.9951	0.0576	0.9955	0.0297	0.3773	0.0518	0.9237
0.065	0.0596	0.9979	0.0579	0.9963	0.0309	0.4651	0.0564	0.9875
0.070	0.0596	0.9979	0.0579	0.9963	0.0324	0.5540	0.0579	0.9963
0.080	0.0596	0.9979	0.0579	0.9963	0.0378	0.6429	0.0579	0.9963
0.090	0.0596	0.9979	0.0579	0.9963	0.0427	0.7535	0.0579	0.9963
0.100	0.0596	0.9979	0.0579	0.9963	0.0478	0.8647	0.0579	0.9963

E PER- α MAIN RESULTS

Table 10 gives the full per- α FDR and recall for the `selfevals` scoring method (the paper default), the disaggregated view behind Figure 2(a,d) and Table 1.